

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 02: Friday 14 August 2026

(Week 1)

Remark. We will discuss the solutions of the following questions on the Friday class. Please try these questions before the class.

Q01.9. Let $f(x) = x^2$. Find $f'(x)$.

Q01.10. Let $f(x) = x^3 + x^2 \tan x$. Find $f'(x)$.

Q01.11. Differentiate $\frac{x^2}{\ln x}$ with respect to x .

Q01.12. Let $y = \tan(t^3 + 2t)$. Find $\frac{dy}{dt}$.

Remark. The above 4 questions are leftover exercises from Tuesday's class.

Q02.1. Consider the curve $2x^3 + y^3 = 5xy$ in $\mathbb{R}^2$. Find $\frac{dy}{dx}$ (in terms of x and y).

Q02.2. Derive the formula $\frac{d}{dx}(\tan^{-1}(x)) = \frac{1}{1+x^2}$.

Q02.3. Let $f(x) = \tan^{-1}(x)$. Find $f''(x)$.

Q02.4. Recall from **Q02.1** the curve $C : 2x^3 + y^3 = 5xy$ in $\mathbb{R}^2$.

(a) Find an equation of the tangent line to the curve C at the point $(1, 2)$.

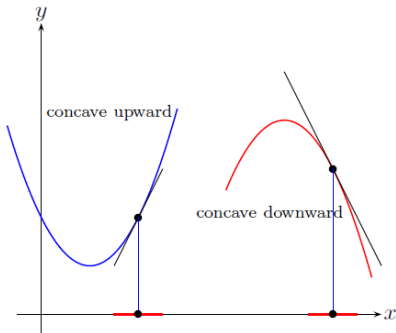
(b) Find an equation of the normal line to the curve C at the point $(1, 2)$.

Q02.5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(1) = 2$ and $f'(1) = 3$. Find an approximate value of $f(1.05)$.

Q02.6. (a) Give a sketch of the graph of a function $f(x)$ defined on an interval I such that $f'(x) < 0$ for all $x \in I$.

(b) Give a sketch of the graph of a function $g(x)$ defined on an interval I such that $g'(x) > 0$ for all $x \in I$.

Q02.7. Suppose a function f is such that $f''(x) < 0$ for all x in an interval. Which of the following two curves can be the graph of $f(x)$? Blue curve or Red curve?



Q02.8. Consider the function $f(x) = x^3 - 3x$.

(a) Find the critical points of $f(x)$.

(b) Find the local maximum point(s) and local minimum point(s) of $f(x)$. Find also the corresponding local maximum value(s) and local minimum value(s) of $f(x)$.

Q02.9. Consider the function

$$f(x) = x^2 e^x, \quad x \in [-3, 2].$$

(i) Find the critical points of $f(x)$ on the interval $[-3, 2]$.

(ii) Find the absolute maximum value of $f(x)$ on the interval $[-3, 2]$.

Q02.10. Find $\lim_{x \rightarrow 1} \frac{e^x - e}{x^2 - 1}$.

Q02.11. Find $\lim_{x \rightarrow \infty} \frac{x^3 + x}{e^x + x^2}$.

Q02.12. Find an antiderivative of the function $f(x) = \cos x$.

Q02.13. Find the following indefinite integrals:

(i) $\int \sin x \, dx$, (ii) $\int x^2 \, dx$, (iii) $\int (2x + 3)^4 \, dx$.

Q02.14. Find $\int x \cos(x^2 + 1) dx$.

Q02.15. Find $\int \frac{e^{3x}}{\sqrt{2e^{3x} + 5}} dx$.

Q02.16. Find $\int x \ln x dx$.

Q02.17. Evaluate $\int_1^e \frac{\ln x}{x} dx$.

Q02.18. Evaluate $\int_0^{\frac{\pi}{2}} x \cos x dx$.

Q02.19. Find $\frac{d}{dx} \int_1^{\sin x} \sqrt{1+t^4} dt$.